LAB ASSIGNMENT 2

Individualized Student Task: Unique Matrix Value Investigation

Student ID (last two digits): 36
d1 = 6 **d2 =** 3

1. Create Your Personal Matrix

Given: d1 = 6 and d2 = 3

Matrix A: A = [[d1 + 2, d2 + 1], [2d1, d2 + 2]]

A = [[6 + 2, 3 + 1], [2(6), 3 + 2]]

Therefore, A = [[8, 4], [12, 5]]

2. Analyze Matrix A

NumPy Code:

```
import numpy as np

d1 = 6
d2 = 3

A = np.array([
    [d1 + 2, d2 + 1],
    [2 * d1, d2 + 2]
])

print("Matrix A:")
print(A)

print("Shape:", A.shape)
print("Determinant:", np.linalg.det(A))
print("Rank:", np.linalg.matrix_rank(A))
print("Eigenvalues:", np.linalg.eigvals(A))

if np.linalg.det(A) != 0:
    print("Inverse:")
    print(np.linalg.inv(A))
```

Results:

Property	Matrix A
Matrix	[[8, 4], [12, 5]]
Shape	(2, 2)
Determinant	-8
Rank	2

Eigenvalues	13.5887, -0.5887
Inverse	[[-0.625, 0.5], [1.5, -1]]

Determinant calculation: $\det(A) = (8\times5) - (4\times12) = 40 - 48 = -8$.

Since the determinant is non-zero, Matrix A is invertible and its rank is 2.

The eigenvalues are approximately 13.5887 and −0.5887.

3. Value Change Experiment

Exactly one value is changed by adding +1. The element A[0,0] is changed from 8 to 9.

Therefore, Matrix B = [[9, 4], [12, 5]].

4. Re-analyze Matrix B

```
B = A.copy()
B[0, 0] += 1

print("Matrix B:")
print(B)

print("Shape:", B.shape)
print("Determinant:", np.linalg.det(B))
print("Rank:", np.linalg.matrix_rank(B))
print("Eigenvalues:", np.linalg.eigvals(B))

if np.linalg.det(B) != 0:
    print("Inverse:")
    print(np.linalg.inv(B))
```

Results:

Property	Matrix B
Matrix	[[9, 4], [12, 5]]
Changed element	A[0,0]: 8 → 9
Shape	(2, 2)
Determinant	-3
Rank	2
Eigenvalues	14.2111, -0.2111
Inverse	[[-1.6667, 1.3333], [4, -3]]

Determinant calculation: $\det(B) = (9\times5) - (4\times12) = 45 - 48 = -3$.

Since the determinant is non-zero, Matrix B is also invertible and its rank is 2.

The eigenvalues are approximately 14.2111 and −0.2111.

5. Comparison & Explanation

1. How did the determinant change and why?

The determinant changed from −8 for Matrix A to −3 for Matrix B. This happened because one element, A[0,0], was increased from 8 to 9, which changed the determinant calculation.

2. Did the rank change? Explain.

No. The rank remained 2 for both matrices. Both determinants are non-zero, so both 2×2 matrices have full rank.

3. How did the eigenvalues respond to the value change?

The eigenvalues changed from approximately 13.5887 and −0.5887 in A to 14.2111 and −0.2111 in B. Thus, changing one matrix value affected both eigenvalues.

4. Is B easier or harder to invert than A? Why?

Both A and B are invertible because their determinants are non-zero. However, B has determinant −3, which is closer to zero than A's determinant −8. Therefore, B is somewhat more sensitive in numerical inversion.

Final Comparison

Property	Matrix A	Matrix B
Matrix	[[8, 4], [12, 5]]	[[9, 4], [12, 5]]
Changed element	—	A[0,0]: 8 → 9
Shape	(2,2)	(2,2)
Determinant	-8	-3
Rank	2	2
Eigenvalues	13.5887, -0.5887	14.2111, -0.2111
Invertible	Yes	Yes